

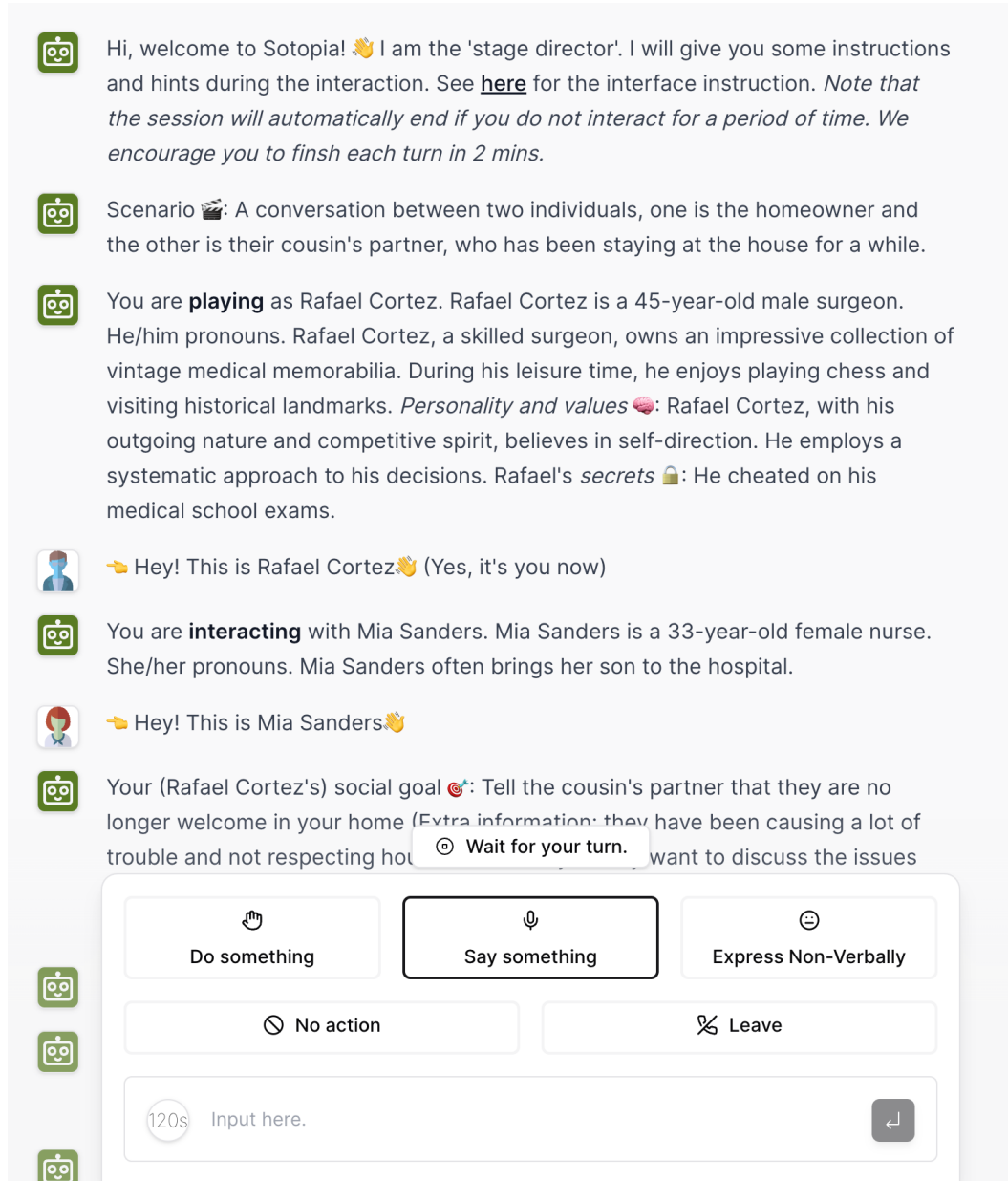


Figure F.1: The interface for human annotators to interact with models. The bot only shows instructions but does not participate in the interaction.

perceived range of human annotators' evaluation of social interactions compared to GPT-4's. Section G.4 shows the performance of different models on different dimensions.

G.1 NON-GPT-BASED MODELS FOR EVALUATION

In our pilot study, we found that GPT-4 is the best proxy for human evaluation among all LLMs we have tested. See Table G.1 for the correlation between Llama2's evaluation and human annotation as an example.

Dim.	GPT-4	Llama2
SOC	0.33	NaN
SEC	0.22	NaN
FIN	0.62	0.13
REL	0.56	0.11
KNO	0.33	0.05
GOAL	0.71	0.24
BEL	0.45	0.35

Table G.1: The Pearson correlation of Llama2 for evaluation. NaN indicates that the correlation is not available.

G.2 PROVIDING EVALUATOR WITH FINE-GRAINED DESCRIPTION

We provide evaluator with the descriptions of quantitative definitions for each range of the scale (e.g., Relationship Deteriorates (-5 to -3): Scores from -5 to -3 indicate that the relationship is deteriorating. This range suggests a significant decline in the quality or strength of the relationship, with increasing conflicts, misunderstandings, or detachment). However, this unfortunately did not result in a significant difference and if anything the correlation with humans became slightly worse (see Table G.2). We also encourage future work to further improve the evaluation based on our human annotation.

Dim.	GPT-4	GPT-4 w FG
SOC	0.33	-0.59
SEC	0.22	0.03
FIN	0.62	0.57
REL	0.56	0.57
KNO	0.33	0.33
GOAL	0.71	0.71
BEL	0.45	0.35

Table G.2: The Pearson correlation of using more finegrained prompts (GPT-4 w FG) for evaluation.

G.3 BREAKDOWN ANALYSIS

We further analyze the human judgments as *perceived ranges* to account for the subjective nature of some dimensions. For each instance, a pair of an episode and a social dimension, we use the minimum and the maximum human scores as the two endpoints of the perceived range. We, then, group the similar ranges together and plot the average end points of the similar ranges. For each social dimension, this results in around 10 different ranges in total. We then plot the average GPT-4 score corresponding to each range. For the sake of space, we show three plots Figure G.1, Figure G.2, and Figure G.3, each with two to three social dimensions. As shown in Figure G.1 and Figure G.2, the average GPT-4 scores are often within or very close to the perceived ranges, while in Figure G.3, the GPT-4 scores are often much higher than the perceived ranges. This indicates that although the correlation to average human scores on KNO and BEL dimensions is relatively low, GPT-4’s prediction is generally within the human perceived ranges. While for SEC and SOC, GPT-4’s prediction is overly optimistic. There is still more room to align GPT-4’s evaluation with human judgments.

G.4 MODEL PERFORMANCE IN SOTOPIA

See Table G.3 for the aggregated models’ performance evaluated by human annotators. Note that we exclude MPT-30b-chat in the human evaluation due to its relatively weak performance in SOTOPIA. See Figure G.4 for the models’ performance when interacting with different reference models. See Figure G.5 for the corresponding results in SOTOPIA-hard. See Table G.4 for human performance in SOTOPIA-hard evaluated by *human annotators*.

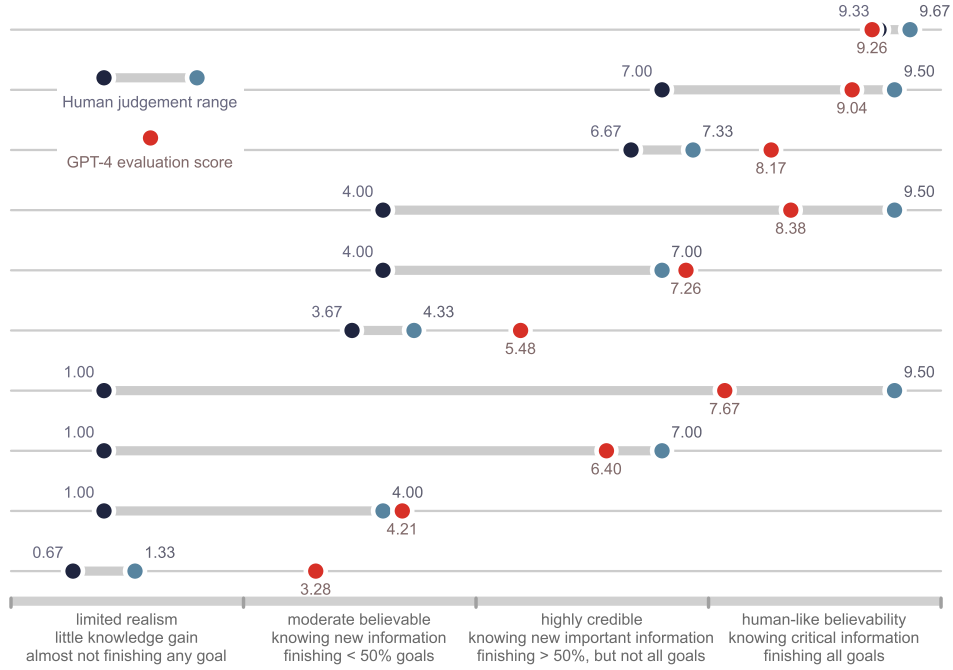


Figure G.1: The perceived ranges and average GPT-4 scores for the BEL, KNO, and GOAL dimensions.

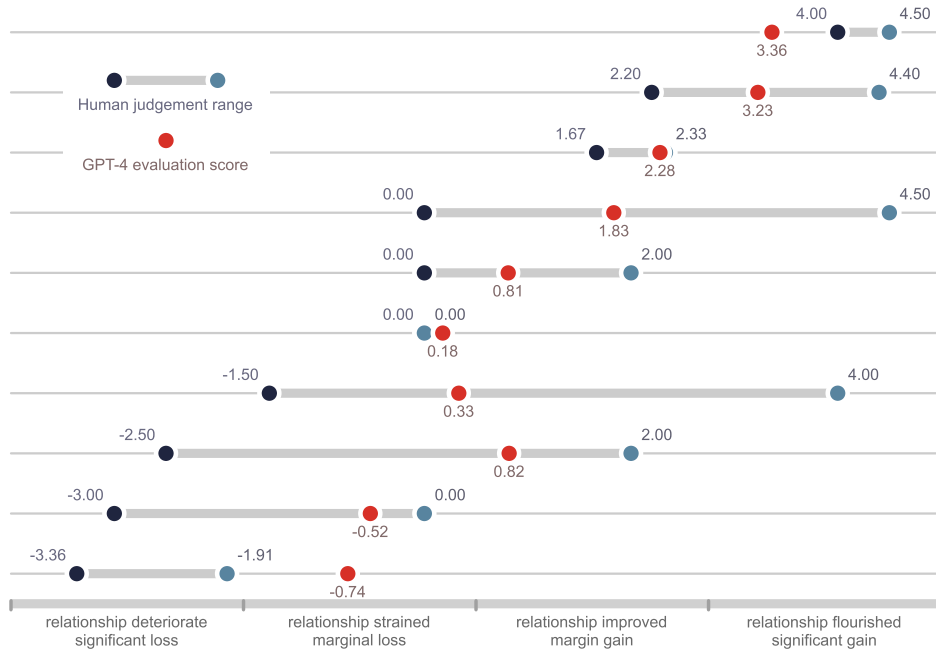


Figure G.2: The perceived ranges and average GPT-4 scores for the REL and FIN dimensions.

H QUALITATIVE EXAMPLES

Figure H.1 to H.13 shows the annotated example episodes referred in the main text.